

Part - A : Conceptual Understanding

① What is Logistic Regression and why is it suitable for classification?

→ Logistic regression is a supervised machine learning algorithm used for classification problems. It predicts the probability of a class.

It is suitable for binary classification problems like High-Risk vs Low-Risk prediction.

② Explain classification performance metrics and why accuracy alone is insufficient.

→ Classification performance metrics include Accuracy, Precision, Recall, F1-Score and ROC-AUC. Accuracy alone is insufficient because it can be misleading when the dataset is imbalanced.

③ Define Type I Error and Type II Error in the context of risk prediction.

→ Type I (FP): A Low-Risk customer is incorrectly classified as High-Risk.

Type II (FN): A High-Risk customer is incorrectly classified as Low-Risk.

Type II Error is more critical because missing risky customers can lead to financial losses.

④ Explain Precision, Recall, F1 score, TPR and FPR

→ (i) Precision : Correct positive predictions out of all positive predictions.

(ii) Recall : Correctly identified actual positives.

(iii) F1 Score : Balance of precision and Recall

(iv) TPR : Same as Recall

(v) FPR : Incorrectly predicted positives among actual negatives.

⑤ What is AUC-ROC and how does it help in evaluating classifiers?

→ ROC curve shows the relationship between TPR and FPR. AUC measures classification performance a higher AUC indicates a better model.

⑥ Why does imbalanced data create problems in classification models?

→ Imbalanced data causes models to favor the majority class and perform poorly on the minority class, leading to incorrect predictions.